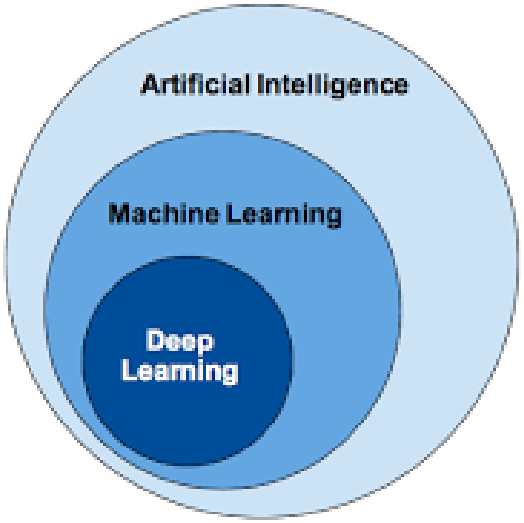
Topic Name: Machine Learning



The concept of learning

X1	X2	Profit
2	1	4000
3	3	9000
1	2	5300
3	2	7100
0	3	6000

X1: # of branches in Amman

X2: # of branches in Dubai

Profit = (X1 + 2 X2) * 1000

relation … equation … model

Prediction

- The importance of learning is to predict the result for new data.

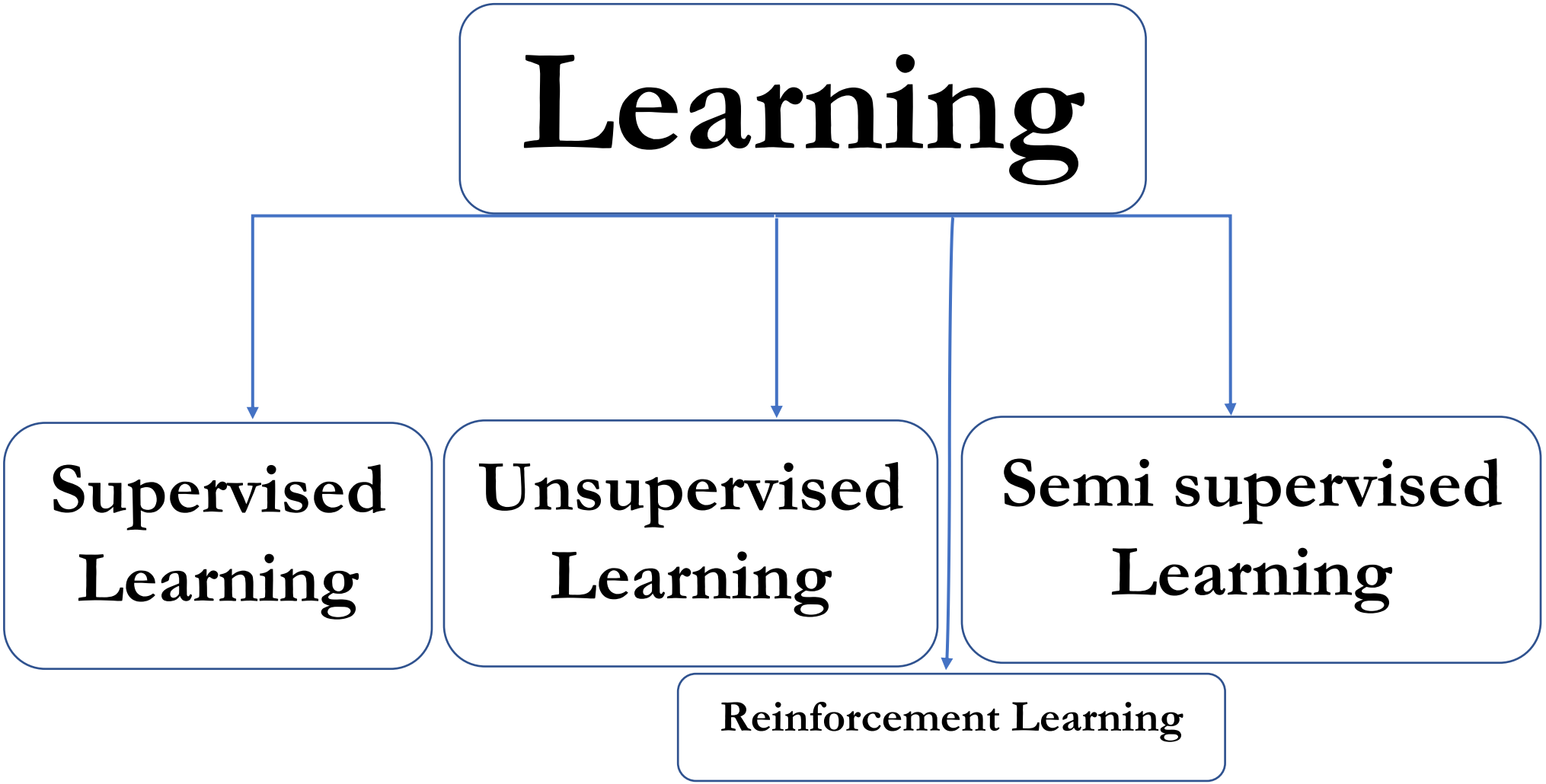
- Example: If someone plans to open 2 branches in Amman, and 3 branches in Dubai. What is the estimated profit according to learned model?

The concept of Features, labels, and samples

Example

age	city	test1	test2	male/female	Cold/Flu
41	Amman	109	30	m	0
22	Salt	124	27	f	0
11	Zarqa	89	32	f	1
57	Madaba	140	37	m	0
44	Amman	99	28	m	1
65	Irbid	112	22	f	0

Types of Learning

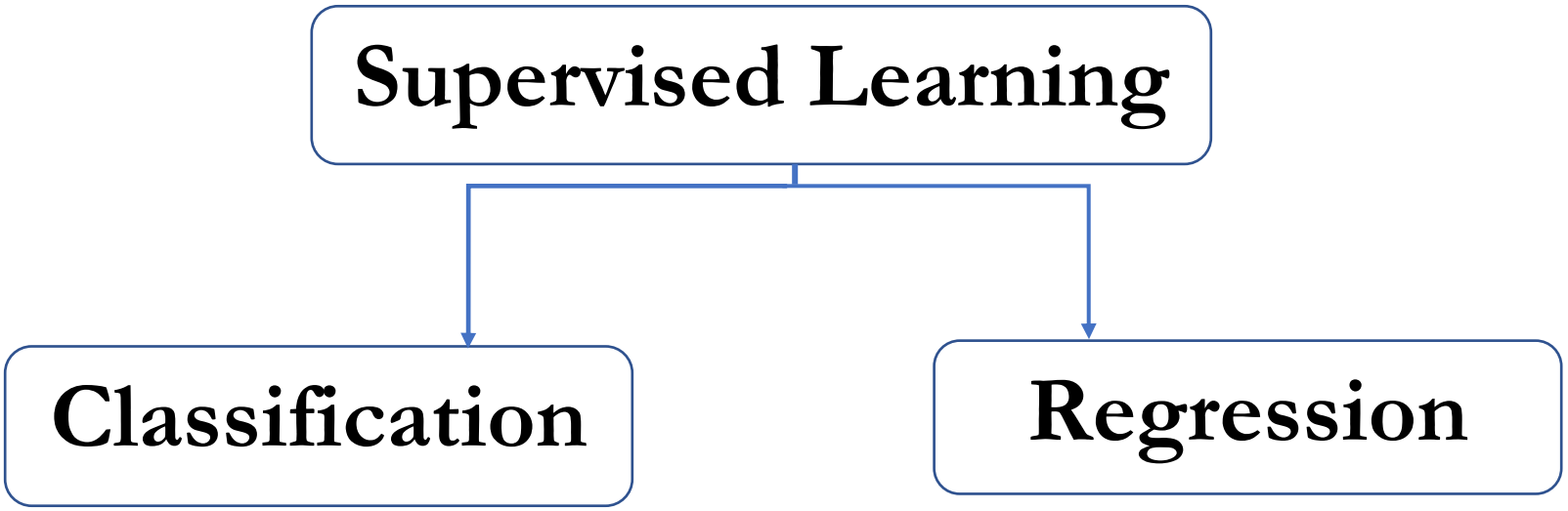


features + (labels/values/outputs)  Supervised learning

features with no (labels/values/outputs)  Unsupervised learning

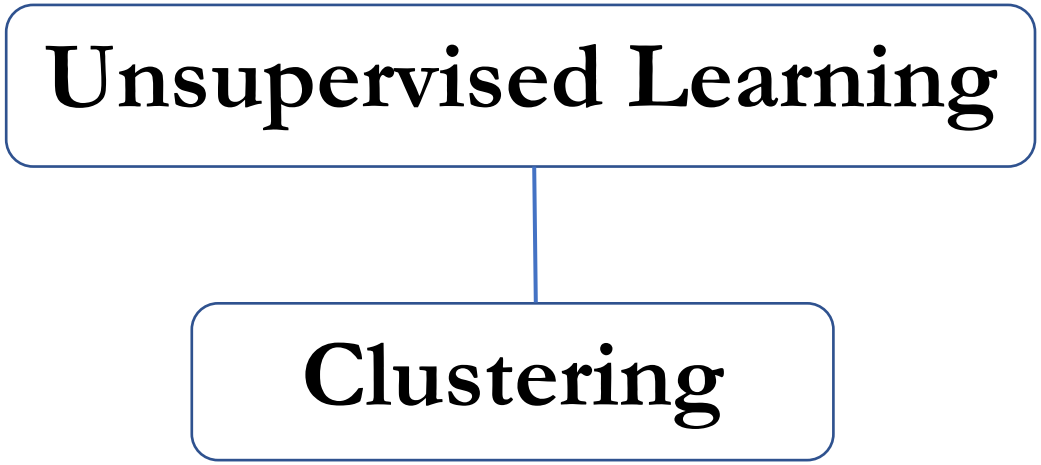
Some samples have labels, and some have not  Semi supervised learning

Classification, Regression, and Clustering



Classification → discrete values

Regression → continuous range


Unsupervised Learning
Clustering

Examples

- **Patients**

Age	city	test1	test2	male/female	Cold/Flu
41	Amman	109	30	m	0
22	Salt	124	27	f	0
11	Zarqa	89	32	f	1
57	Madaba	140	37	m	0
44	Amman	99	28	m	1
65	Irbid	112	22	f	0

Category: classification

- **Pictures of animals**

Width	height	color	background	Object
--	--	--	--	dog
--	--	--	--	cat
--	--	--	--	cat
--	--	--	--	mouse
--	--	--	--	dog
--	--	--	--	mouse

Category: classification

- **Houses**

Area	# of rooms	city	Street No.	Price
--	--	--	--	42,500
--	--	--	--	39,000
--	--	--	--	55,000
--	--	--	--	19,400
--	--	--	--	67,230
--	--	--	--	23,800

Category: regression

Again… The concept of learning

Regression case:

X1	X2	Profit
2	1	4000
3	3	9000
1	2	5300
3	2	7100
0	3	6000

Algorithm-A

Profit = (X1 + 2 X2) * 1000

Algorithm-B

Profit = (X1 + 2 X2) * 1000 + 200

Classification case

x1	x2	x3	Y
2	4	1	1
7	5	5	0
4	6	2	0
3	9	6	1
11	2	7	1

- Algorithm-A gives the following model:

$$Y = \begin{cases} 0 \text{ , if } x1 + x2 + x3 - 2 < 15 \\ 1 \text{ , otherwise} \end{cases}$$

- Algorithm-B gives the following model:

$$Y = \begin{cases} 0 \text{ , if } x1 + x2 + x3 < 20 \\ 1 \text{ , otherwise} \end{cases}$$